

Lecture 8: ICEnet and Regularization

Deep Learning for Actuarial Modeling

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ABSTRACT

Regularization is a model fitting technique that aims at variable shrinkage and variable selection during their estimation procedure. This can be achieved by penalizing extreme parameter values, making models generally more smooth (in some sense). This lecture discusses different regularization methods, and then presents the ICEnet architecture of Richman and Wüthrich (2024), which constrains neural-network predictions so that they are smooth and monotone with respect to selected risk factors, while keeping the deep-learning framework.

1 Regularization

OVERVIEW

Regularization is a model fitting technique that aims at variable shrinkage and variable selection during their estimation procedure. This can be achieved by penalizing extreme parameter values, making models generally more smooth (in some sense). This section discusses different regularization methods.

1.1 Regularization: introduction

- Generally speaking, a regression model $\mathbf{X} \mapsto \mu(\mathbf{X})$ is correctly specified if it integrates all relevant covariates in the correct form, and if there is no redundant, missing or wrong covariate in the regression function:
 - *redundant* means that we include multiple covariates that essentially present the same information (e.g., being collinear);
 - *missing* means that we have forgotten important covariate information that would be available; and
 - *wrong* means that the covariate does not impact the response.
- In real world problems, with unknown regression functions, this is part of covariate pre-processing, model selection and covariate selection.
- One might be tempted to include any available information into the regression model, to ensure that nothing gets forgotten. However, fitting on finite samples, this is usually not a good strategy, because one may easily run into over-fitting issues, difficulties with collinearity in covariates, in identifiability issues, and, generally, in a poor predictive model because model fitting has not been successful.
- Model complexity control is crucial in a successful model fitting and predictive modeling procedure. Typically, one aims for a *sparse* model, meaning that one aims for *parsimony* having the fewest necessary variables but still providing accurate predictions.
- In practice, neither knowing the true model nor the (causal) regression structure and the factors that impact the responses, one often starts with slightly too large models, that one tries to shrink by regularization. Regularization penalizes model complexity and/or extreme regression coefficients.

1.2 Regularization techniques

We describe the most popular regularization techniques in this section; see Hastie, Tibshirani and Wainwright (2015).

- *Ridge regularization* (also known as Tikhonov (1943) regularization or L^2 -regularization);
- *LASSO regularization* of Tibshirani (1996) (L^1 -regularization);
- *Elastic net regularization* of Zou and Hastie (2005);
- *Fused LASSO regularization* of Tibshirani et al. (2005);
- *Group LASSO regularization* of Yuan and Lin (2006);
- *Smoothly clipped absolute deviation (SCAD) regularization* of Fan and Li (2001); we skip the details of SCAD in this notebook.

1.3 Regularization preparation

- Consider a parametric regression estimation $\mu_{\hat{\vartheta}}(\cdot)$ by selecting from a class of candidate models $\mathcal{M} = \{\mu_{\vartheta}\}_{\vartheta}$ being parametrized by ϑ .
- The parameter ϑ is assumed to be a $(r + 1)$ -dimensional vector

$$\vartheta = (\vartheta_0, \vartheta_1, \dots, \vartheta_r)^\top \in \mathbb{R}^{r+1}.$$

Typically, $(\vartheta_k)_{k=1}^r$ parametrizes the terms in $\mu_{\vartheta}(\mathbf{X})$ involving the covariates $\mathbf{X}$, and ϑ_0 is a parameter for the covariate-free part of the regression function determining the overall level of the regression function; ϑ_0 is the *intercept/bias term*, see GLM above.

- This bias term ϑ_0 needs to be excluded from regularization to be able to calibrate the overall level of the model.
- Denote by $\vartheta_{\setminus 0} = (\vartheta_1, \dots, \vartheta_r)^\top \in \mathbb{R}^r$ the parameter vector excluding the bias term ϑ_0 .

1.4 Regularization: penalty function

- Regularized parameter estimation is achieved by

$$\widehat{\vartheta} \in \arg \min_{\vartheta} \left(\sum_{i=1}^n \frac{v_i}{\varphi} L(Y_i, \mu_{\vartheta}(\mathbf{X}_i)) + \lambda R(\vartheta_{\setminus 0}) \right),$$

where $R : \mathbb{R}^r \rightarrow \mathbb{R}_+$ is a *penalty function* and $\lambda \geq 0$ is the *regularization parameter*.

- We give the most common examples of penalty functions, below.
- Noteworthy, we do not consider any scaling $1/n$ in front of the sum. This scaling does not affect the optimal solution in the non-regularized case, but it makes the regularization parameter λ sample size independent.

1.5 Ridge regularization/regression

- Ridge regression selects a squared L^2 -norm penalty function for R

$$\widehat{\vartheta}^{\text{ridge}} \in \arg \min_{\vartheta} \left(\sum_{i=1}^n \frac{v_i}{\varphi} L(Y_i, \mu_{\vartheta}(\mathbf{X}_i)) + \lambda \|\vartheta_{\setminus 0}\|_2^2 \right).$$

- Ridge regularization generally punishes large values in $(\vartheta_k)_{k=1}^r$, this is called *shrinkage*.
- The level of shrinkage is determined by the regularization parameter $\lambda \geq 0$, and an optimal choice is determined, e.g., by cross-validation.
- This intuition of shrinkage indicates why we exclude the bias term ϑ_0 from regularization. Otherwise, we receive a statistically biased model.

1.6 LASSO regularization/regression

- LASSO regularization (least absolute shrinkage and selection operator regularization) selects an L^1 -norm penalty function for R

$$\hat{\vartheta}^{\text{LASSO}} \in \arg \min_{\vartheta} \left(\sum_{i=1}^n \frac{v_i}{\varphi} L(Y_i, \mu_{\vartheta}(\mathbf{X}_i)) + \lambda \|\vartheta_{\setminus 0}\|_1 \right).$$

- The behavior of LASSO regularization is fundamentally different from ridge regularization. This difference results from the fact that the L^2 -norm is differentiable in the origin and the L^1 -norm is not.

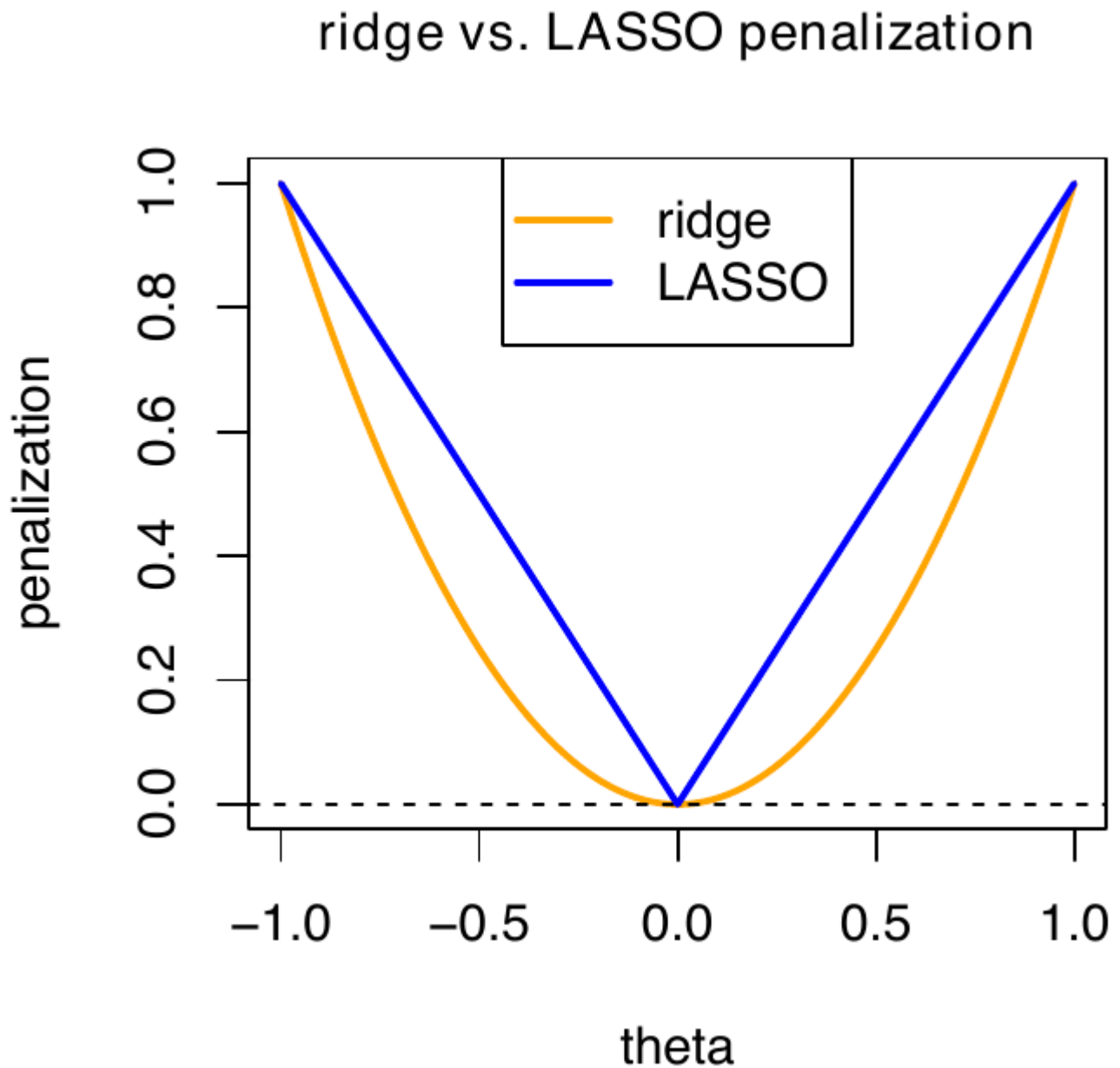


Figure 1: Ridge versus LASSO penalization.

- The consequence of this degeneracy of the L^1 -norm in the origin is that some of the components of $(\vartheta_k)_{k=1}^r$ may be optimal in LASSO regularization if they are *exactly* equal to zero.

- The more we increase the regularization parameter $\lambda \uparrow \infty$, the more components of $(\vartheta_k)_{k=1}^r$ are optimal in precisely setting them to zero.
- Increasing the regularization parameter λ leads to sparsity of non-zero values in $(\vartheta_k)_{k=1}^r$, and, consequently, there results a *sparse regression model* for large regularization parameters $\lambda \gg 0$.
- This can be interpreted as *variable importance*, i.e., for large λ only important covariate components survive.
- Often, one performs variable selection with LASSO regularization, and then one fits a non-regularized model on the selected components.

1.7 Geometric interpretation of ridge vs. LASSO

- Generally, these regularized regression problems can be interpreted geometrically using the method of Lagrange:
 - the feasible set of solutions in the ridge regression case is a L^2 -ball, and
 - in the LASSO regression case a L^1 -cube;

we refer to the next graph.

- This geometric difference (having corners or not) precisely distinguishes sparsity of LASSO from shrinkage of ridge regression, i.e., this is related to the difference of these convex geometric sets having differentiable boundaries or not.
- In fact, one considers orthogonal projections to the ball and cube, respectively, and sparsity in LASSO results from corner solutions in the cube case.

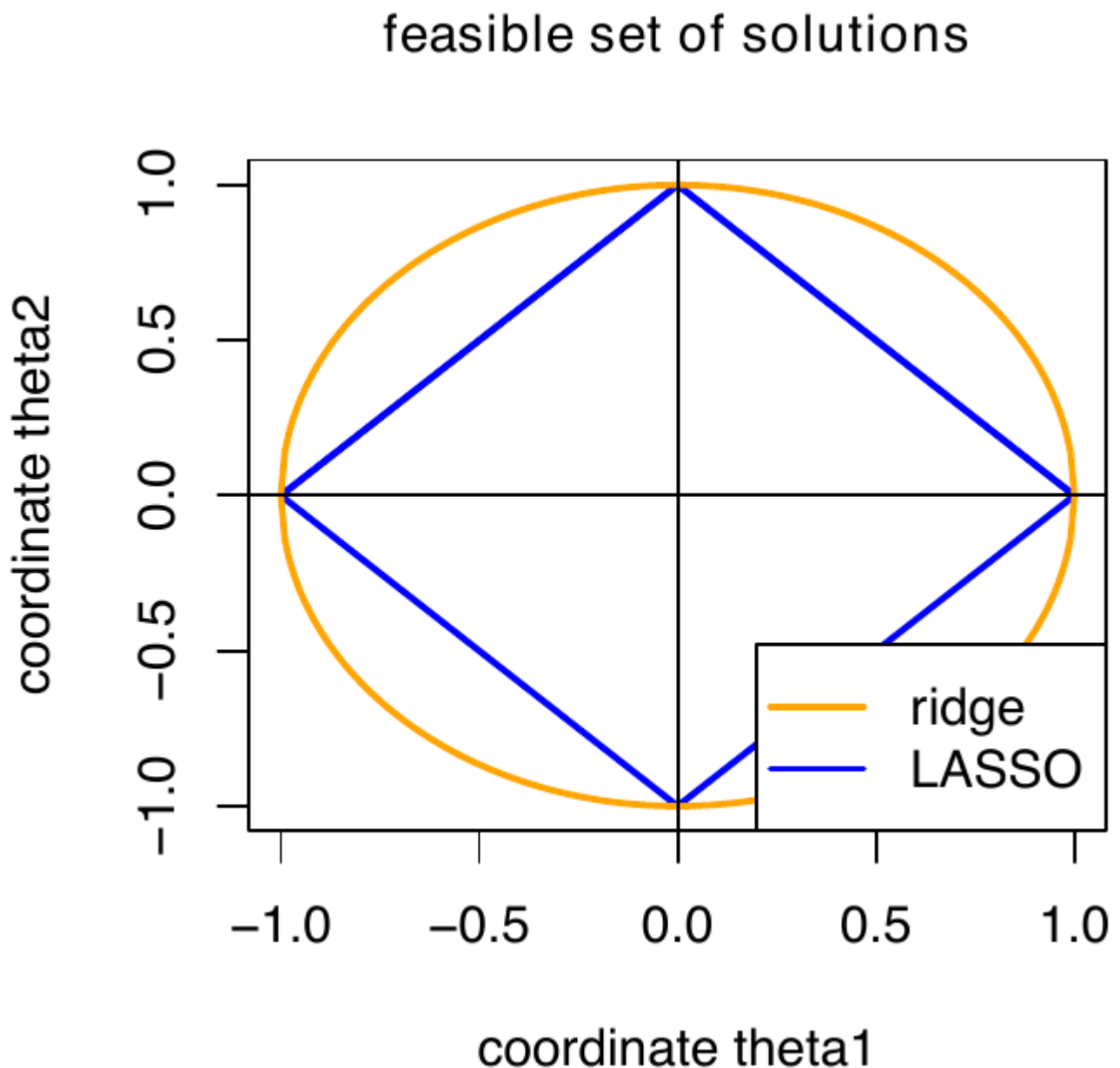


Figure 2: Feasible set of solutions: the L^2 -ball of ridge regression and the L^1 -cube of LASSO regression.

1.8 Best-subset selection regularization

- *Best-subset selection regularization* selects a L^0 -norm penalty function for R

$$\hat{\vartheta}^{\text{BSS}} \in \arg \min_{\vartheta} \left(\sum_{i=1}^n \frac{v_i}{\varphi} L(Y_i, \mu_{\vartheta}(\mathbf{X}_i)) + \lambda \sum_{k=1}^r \mathbf{1}_{\{\vartheta_k \neq 0\}} \right).$$

- This version is not used very often in applications mainly because optimizing the above is difficult, in fact, LASSO regularization is considered as a tractable version

instead.

1.9 Elastic regularization

- *Elastic net regularization* combines LASSO and ridge regularization

$$\widehat{\vartheta}^{\text{net}} \in \arg \min_{\vartheta} \left(\sum_{i=1}^n \frac{v_i}{\varphi} L(Y_i, \mu_{\vartheta}(\mathbf{X}_i)) + \lambda (\alpha \|\vartheta_{\setminus 0}\|_1 + (1 - \alpha) \|\vartheta_{\setminus 0}\|_2^2) \right),$$

with $\alpha \in [0, 1]$.

- Elastic net regularization overcomes some issues of LASSO, e.g., LASSO has the tendency to group effects by assigning similar weights to correlated covariate components.

1.10 Grouped covariates

- The previous regularization proposals have been focusing on shrinking the regression parameters $(\vartheta_k)_{k=1}^r$ and/or to make regression models sparse by setting some of these parameters ϑ_k exactly to zero.
- In these regularization methods all terms and parameters $(\vartheta_k)_{k=1}^r$ are considered individually.
- If we want to treat some of them jointly in a group, e.g., for a dummy coded categorical covariate, regularization should act simultaneously on all parameters that belong to that group.
- We group the (categorical) covariates. Assume we have G groups

$$\vartheta_{\setminus 0} = \left(\vartheta^{(1)}, \dots, \vartheta^{(G)} \right)^{\top} \in \mathbb{R}^{d_1 + \dots + d_G} = \mathbb{R}^r,$$

where each group $\vartheta^{(k)} \in \mathbb{R}^{d_k}$ contains d_k components of $\vartheta_{\setminus 0}$.

1.11 Group LASSO regularization

- *Group LASSO regularization* is obtained by solving

$$\widehat{\vartheta} \in \arg \min_{\vartheta} \left(\sum_{i=1}^n \frac{v_i}{\varphi} L(Y_i, \mu_{\vartheta}(\mathbf{X}_i)) + \sum_{k=1}^G \lambda_k \|\vartheta^{(k)}\|_2 \right),$$

for regularization parameters $\lambda_k \geq 0$.

- For increasing regularization parameters λ_k , group LASSO regularization leads to sparsity in setting the entire block of covariates $\vartheta^{(k)} \in \mathbb{R}^{d_k}$ simultaneously equal to zero.

1.12 Fused LASSO regularization

- Assume there is an adjacency relation on the covariate components, i.e., the covariate component X_j is naturally embedded between the components X_{j-1} and X_{j+1} , and we try to achieve a certain smoothness of the regression parameters for these covariates.
- *Fused LASSO regularization* is obtained by solving

$$\widehat{\vartheta} \in \arg \min_{\vartheta} \left(\sum_{i=1}^n \frac{v_i}{\varphi} L(Y_i, \mu_{\vartheta}(\mathbf{X}_i)) + \sum_{j=2}^r \lambda_j |\vartheta_j - \vartheta_{j-1}| \right),$$

for regularization parameters $\lambda_j \geq 0$.

- The fused LASSO proposal enforces sparsity in the non-zero components, but also sparsity in the different regression parameter values for adjacent variables. It considers first order differences, and one could also consider second (or higher) order differences.

1.13 Positivity and monotonicity regularization

- One may want to enforce that the parameters are positive

$$\widehat{\vartheta} \in \arg \min_{\vartheta} \left(\sum_{i=1}^n \frac{v_i}{\varphi} L(Y_i, \mu_{\vartheta}(\mathbf{X}_i)) + \sum_{j=1}^r \lambda_j (0 - \vartheta_j)_+ \right).$$

- Enforcement of a monotone increasing property

$$\widehat{\vartheta} \in \arg \min_{\vartheta} \left(\sum_{i=1}^n \frac{v_i}{\varphi} L(Y_i, \mu_{\vartheta}(\mathbf{X}_i)) + \sum_{j=1}^r \lambda_j (\vartheta_{j-1} - \vartheta_j)_+ \right),$$

with *rectified linear unit (ReLU) function*

$$u \mapsto (u)_+ = \max\{u, 0\} = u \mathbf{1}_{\{u>0\}}.$$

- The above regularizes parameters $\vartheta_{\setminus 0}$, but one could also enforce smoothness and monotonicity in the regression function components (e.g., mortality graduation); see Richman and Wüthrich (2024).

2 ICEnet: smoothness and monotonicity constraints for neural networks

OVERVIEW

This section presents the ICEnet architecture introduced by Richman and Wüthrich (2024). The central idea is to constrain neural-network predictions so that they are smooth and monotone with respect to selected risk factors, while keeping the deep-learning framework.

The contribution is directly connected to regularization: instead of penalizing only the parameters, ICEnet penalizes the behavior of the predictive function on *Individual*

2.1 Actuarial motivation

- In actuarial pricing, tariff curves are often expected to satisfy structural requirements:
 - *smoothness*, to prevent abrupt premium changes that are difficult to justify to policyholders;
 - *monotonicity*, when actuarial knowledge implies a clear direction of risk.
- In GLMs, these constraints can be imposed by directly modifying the model coefficients.
- In neural networks, operating directly on the model coefficients or architecture to impose these constraints is less feasible.
- ICEnet overcomes this limitation by imposing constraints on the model predictions, rather than on the network parameters themselves.

2.2 ICE and PDP: interpretability ingredients

- Consider the fitted regression function

$$\mathbf{X} = (X_1, \dots, X_q)^\top \mapsto \mu_{\hat{\vartheta}}(\mathbf{X}),$$

where q is the number of covariates.

- For covariate j , define the set of values

$$\mathcal{A}_j = \{a_1^j, \dots, a_{K_j}^j\}.$$

- If X_j is **categorical**, $\mathcal{A}_j$ contains all its K_j possible levels.
- If X_j is **continuous**, $\mathcal{A}_j$ contains K_j representative values selected over its observed range, for example using quantile binning. These values are typically ordered as $a_1^j < \dots < a_{K_j}^j$.

- For each observation i , define the pseudo-input

$$\widetilde{\mathbf{X}}_i^{[j]}(u) = (X_{i,1}, \dots, X_{i,j-1}, a_u^j, X_{i,j+1}, \dots, X_{i,q})^\top, \quad u = 1, \dots, K_j.$$

- The ICE curve for observation i and covariate j is

$$\text{ICE}_i^{[j]} = \left[\mu_{\widehat{\vartheta}} \left(\widetilde{\mathbf{X}}_i^{[j]}(1) \right), \dots, \mu_{\widehat{\vartheta}} \left(\widetilde{\mathbf{X}}_i^{[j]}(K_j) \right) \right]^\top.$$

NOTE

The corresponding Partial Dependence Plot (PDP) is the empirical average of the ICE curves:

$$\text{PDP}^{[j]} = \frac{1}{n} \sum_{i=1}^n \text{ICE}_i^{[j]}.$$

2.3 Key idea of ICEnet

- Select the covariates requiring smoothness $\mathcal{S} \subset \{1, \dots, q\}$ or monotonicity, $\mathcal{M} \subset \{1, \dots, q\}$.
- For each observation $\mathbf{X}_i$, generate pseudo-inputs $\widetilde{\mathbf{X}}_i^{[j]}(u)$ by varying the constrained covariates.
- Apply the same model $\mu_{\widehat{\vartheta}}(\cdot)$ to:
 - observed inputs, to obtain $\widehat{Y}_i$;
 - pseudo-inputs, to obtain ICE curves.
- Fit the model using a compound loss:

predictive loss + smoothness penalty + monotonicity penalty.

2.4 Compound ICEnet loss

For each observation i , the ICEnet uses the compound loss

$$L(\vartheta; i) = L^D(Y_i, \mu_{\vartheta}(\mathbf{X}_i)) + L_2(\mu_{\vartheta}(\mathbf{X}_i)) + L_3(\mu_{\vartheta}(\mathbf{X}_i)),$$

where L^D is the deviance loss associated with the considered regression problem.

- $L_2(\mu_{\vartheta}(\mathbf{X}_i))$ penalizes violations of smoothness for the covariates in $\mathcal{S}$.
- $L_3(\mu_{\vartheta}(\mathbf{X}_i))$ penalizes violations of monotonicity for the covariates in $\mathcal{M}$.
- These penalty terms depend on non-negative regularization parameters λ_{sj} and λ_{mj} , which control the trade-off between predictive accuracy and the strength of the imposed constraints.

2.5 Smoothness penalty

- The ICEnet uses a Whittaker-Henderson type penalty to impose smoothness on the ICE outputs.
- The first-order difference is defined for $u = 2, \dots, K_j$ as

$$\Delta^1 \left(\mu_{\vartheta} \left(\widetilde{\mathbf{X}}_i^{[j]}(u) \right) \right) = \mu_{\vartheta} \left(\widetilde{\mathbf{X}}_i^{[j]}(u) \right) - \mu_{\vartheta} \left(\widetilde{\mathbf{X}}_i^{[j]}(u-1) \right).$$

- For $\tau \geq 2$, the τ -th order difference is defined recursively for $u = \tau + 1, \dots, K_j$ as

$$\Delta^{\tau} \left(\mu_{\vartheta} \left(\widetilde{\mathbf{X}}_i^{[j]}(u) \right) \right) = \Delta^{\tau-1} \left(\mu_{\vartheta} \left(\widetilde{\mathbf{X}}_i^{[j]}(u) \right) \right) - \Delta^{\tau-1} \left(\mu_{\vartheta} \left(\widetilde{\mathbf{X}}_i^{[j]}(u-1) \right) \right).$$

- The ICEnet penalizes squared third-order differences:

$$L_2(\mu_{\vartheta}(\mathbf{X}_i)) = \sum_{j \in \mathcal{S}} \sum_{u=4}^{K_j} \lambda_{s,j} \left[\Delta^3 \left(\mu_{\vartheta} \left(\widetilde{\mathbf{X}}_i^{[j]}(u) \right) \right) \right]^2.$$

2.6 Monotonicity penalty

- Monotonicity is imposed by penalizing first differences with the undesired sign:

$$L_3(\mu_{\vartheta}(\mathbf{X}_i)) = \sum_{j \in \mathcal{M}} \sum_{u=2}^{K_j} \lambda_{m,j} \max \left[\delta_j \Delta^1 \left(\mu_{\vartheta} \left(\widetilde{\mathbf{X}}_i^{[j]}(u) \right) \right), 0 \right].$$

- $\delta_j = -1$ enforces a monotone increasing relationship.
- $\delta_j = +1$ enforces a monotone decreasing relationship.
- If the ICE curve follows the required direction, the penalty is zero.

3 ICEnet example: French MTPL data

- Goal: predict claims frequency.
- Constrained variables:

Covariate	Smoothness	Monotonicity	Direction
Driver age	10	0	-
Vehicle age	1	0	-
Bonus-malus	1	100	increasing
Density	1	100	increasing
Vehicle power	1	100	increasing

- Setup: FCN with depth $d = 3$, layer dimensions (32, 16, 8), ReLU activation, embedding layers for unordered categorical variables and Adam with early stopping.

3.1 Main predictive results

Model	Learn	Test
FCN	0.4500 (0.000399)	0.4511 (0.000663)
FCN nagging	0.4491	0.4504
ICEnet	0.4510 (0.000340)	0.4513 (0.000446)
ICEnet nagging	0.4506	0.4508

- Adding constraints worsens Poisson deviance only marginally.
- The learning-test gap is smaller for ICEnet: the constraints also act as a form of regularization.

3.2 Partial dependence plot

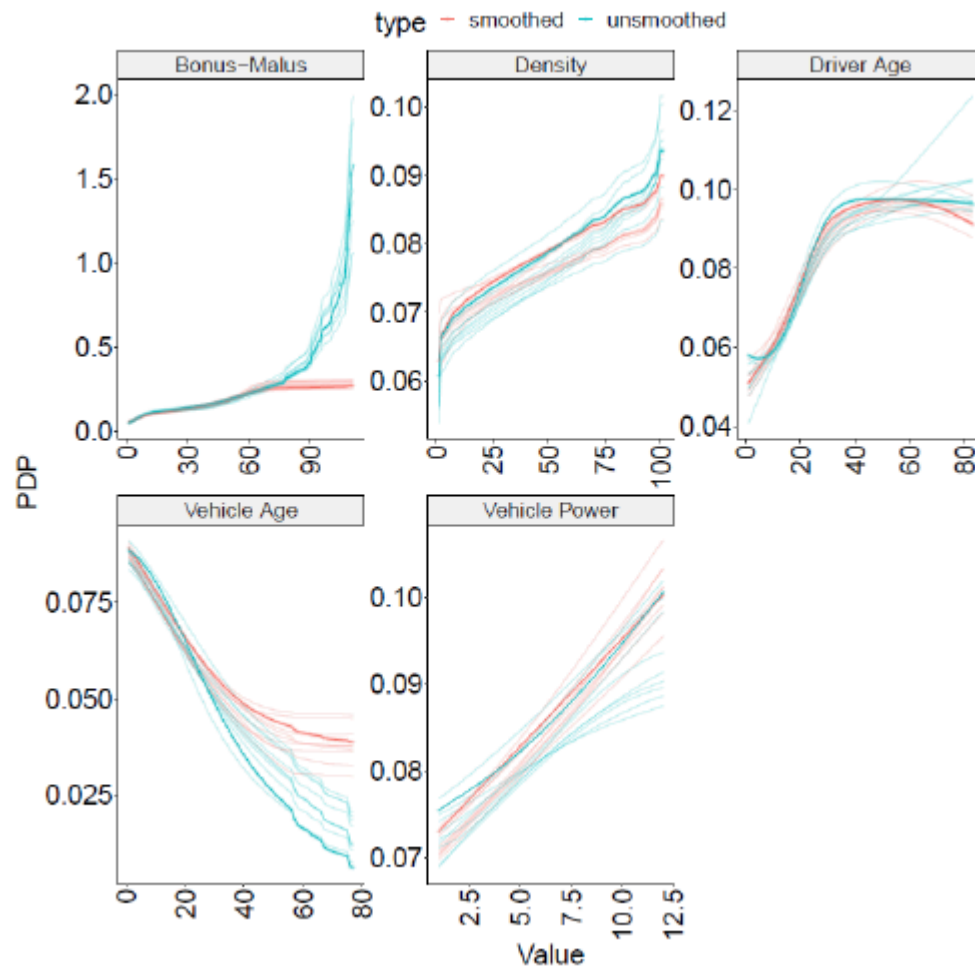


Figure 3: Partial dependence plots for the Bonus-Malus, Density, Driver Age, Vehicle Age, and Vehicle Power covariates for different training attempts.

3.3 ICE plots

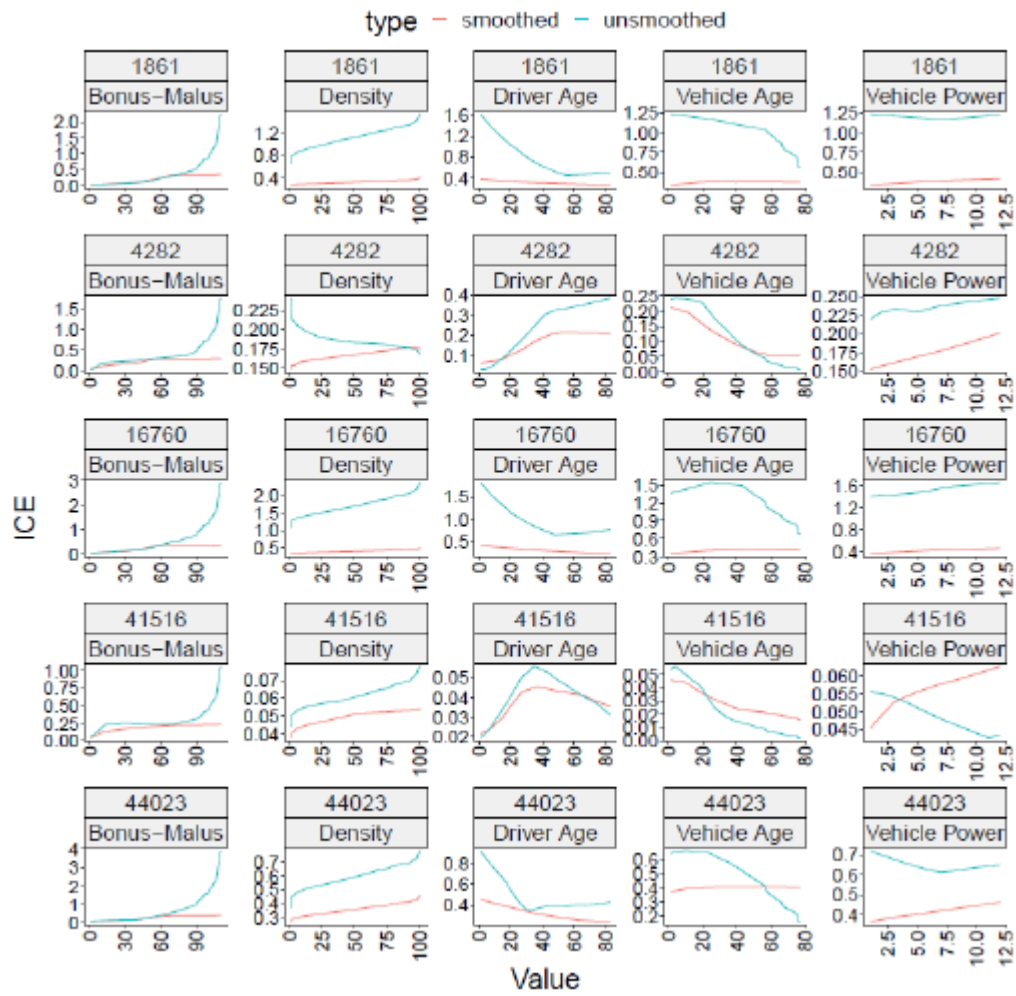


Figure 4: ICE plots of the output of the FCN and the ICENet for instances chosen to be the least monotonic.

3.4 PDPs under different constraint-parameter values

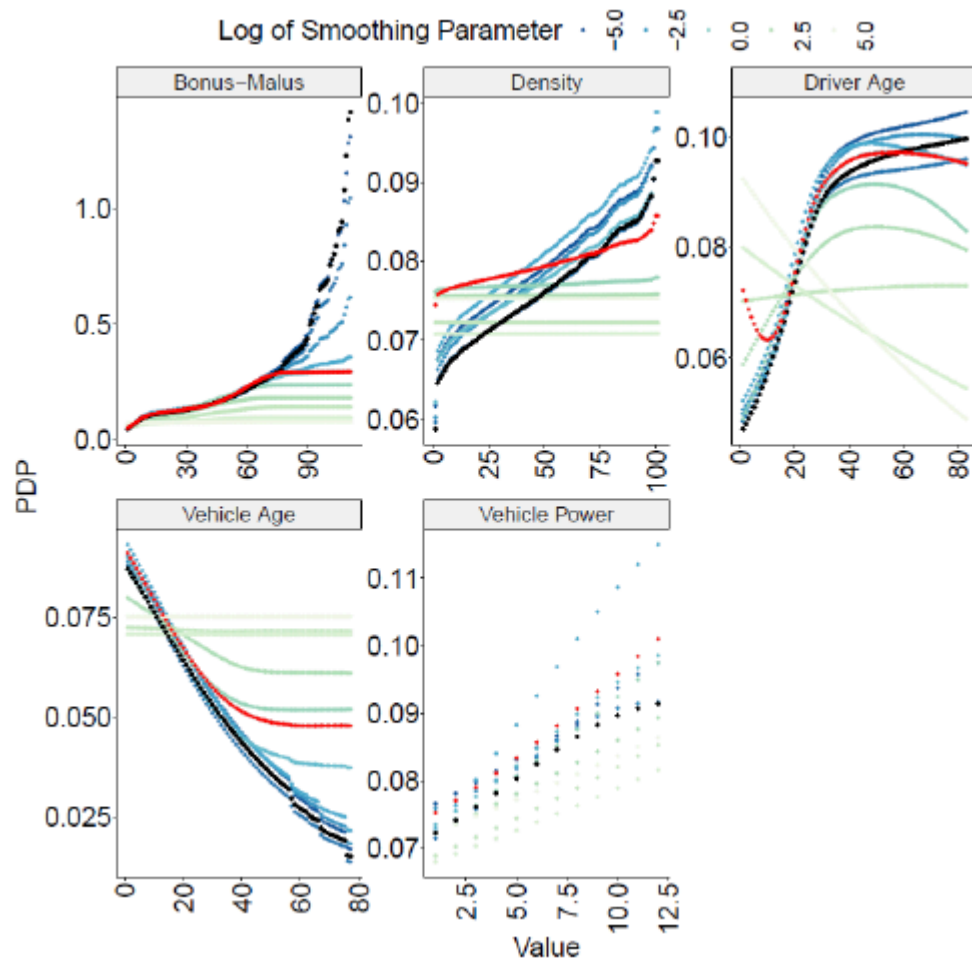


Figure 5: PDPs for constraint parameters ranging from 10^{-5} to 10^5 .

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